

Biotechnology

Bio + technology $\Rightarrow$ Term by Karl Euky (1970)
for large scale production of pigs.
fusion of living and non-living science.

> It is the use of living organism or their ^{like enzymes} component to produce product and services useful to human being.

Two types:

1. Old/Traditional Biotech. $\rightarrow$ Before 1970
2. New/Modern Biotech

Old Biotechnology:

In this natural efficiency of living organisms are used to produce useful product and services
eg: formation of bread, cheese, curd, wine etc.
by microbes

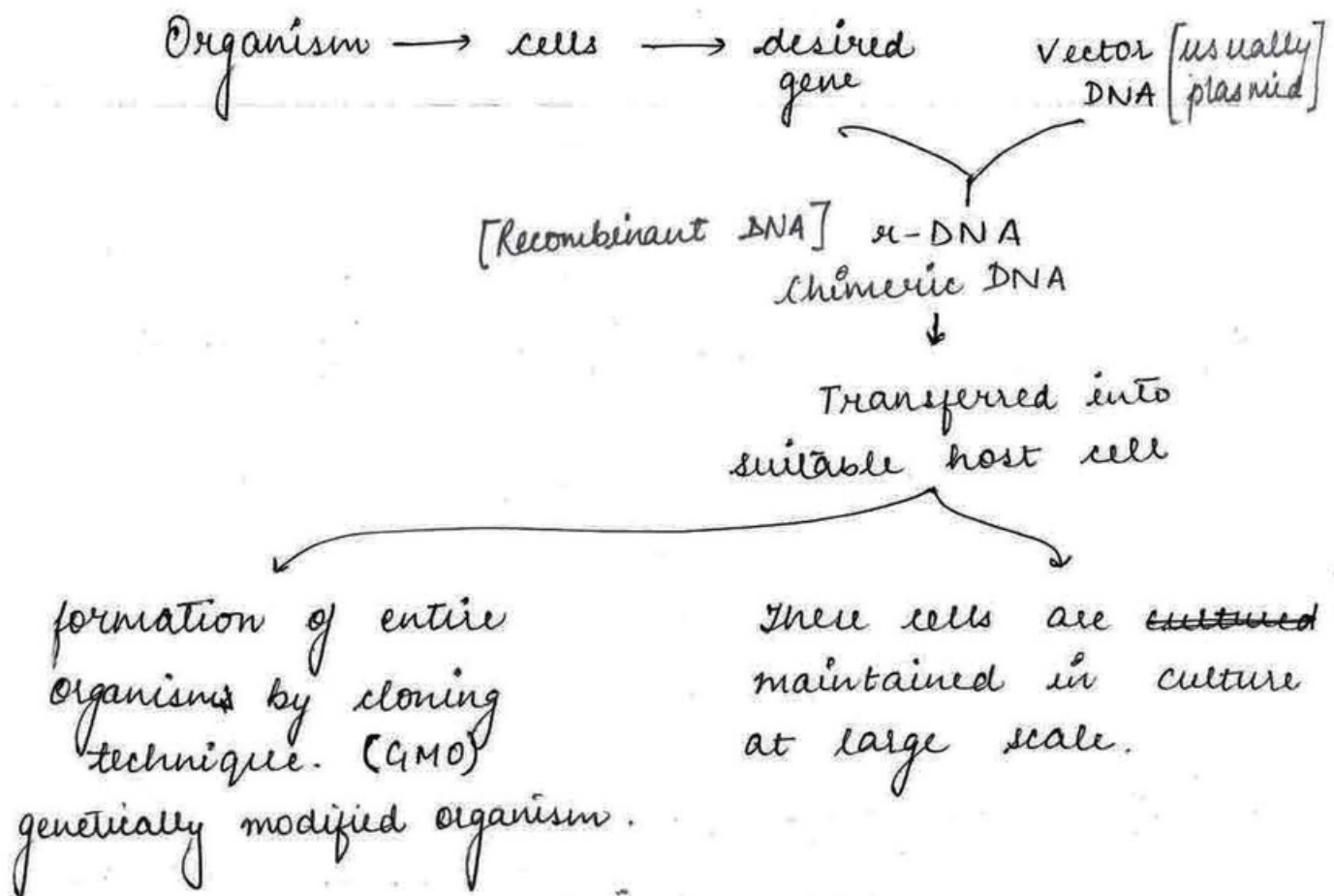
New/Modern Biotechnology

Based on genetic engineering.
It involves 2 main principles

formation of transgenic organism / GMO by technology of genetic engineering.

Maintenance of GMO/
Transgenic organisms.
(Large scale culture to obtain desired products)

By recombinant DNA (rDNA) technology and gene transfer technique



Two main landmark discovery:

- $\Rightarrow$ discovery of restriction endonuclease enzyme
- $\Rightarrow$ discovery of Plasmid autonomous replicating circular extra chromosomal DNA

* Cohen and H. Boyer

set rDNA

'They combined an antibiotic resistance gene of bacteria with the plasmid of Salmonella Typhimurium'

desired gene
vector DNA

Paul Bergh : 1st successful gene transferred.

A gene of SV-40 virus $\xrightarrow[\text{by } \lambda\text{-phage vector}]{\text{to}}$ E. coli

Major steps:

- > Isolation of desired ^{gene} and formation of r-DNA (recombinant) by combining it with vector DNA
- > Transfer of r-DNA into suitable host cell.
- > Selection and screening of recombinant host cell.
- > Maintenance and culturing of recombinant host cell at large scale to obtain desired product.

r-DNA formation and gene transfer Technique

I Tools required: 4 Types

- + Enzymes
- + Vector DNA
- + Passenger / DNA
- + Host cell

I) Enzymes:

several types

⑤ Types

- + lysing enzyme
- + cleaving enzyme
- + synthesizing enz.
- + joining enzy.
- + other enzymes

i) Lysing enzyme

(enzyme for lysis of cells).

eg: lysozyme (for bacteria)

chitinase (for fungi)

cellulase [for plant cell]

Pectinase [for plant cell]

Protease [for animal cell]

Lipase [for animal cell]

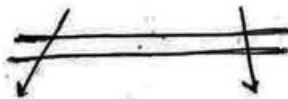
ii) cleaving enzyme

(most imp.)

Enzyme used to cut the DNA

It is of 2 types:

a) Exonuclease



cuts DNA at terminal ends.

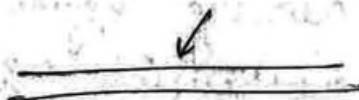
not so used

b) Endonuclease

cuts DNA at intermediate places

commonly used

special type: Restriction endonuclease enz.



* Restriction endonuclease enzyme: [Molecular scissors]
v. v. imp.

- > enzyme which cuts DNA obliquely/straight at specific sites which is known as specific site.
- > Each Restriction enzyme have its own specific site.

1st discovered enz. $\Rightarrow$ Hind II by Smith
from Haemophilous influenzae bacteric

later on, ECORI enzymes from E. coli
(by Smith, Nathans and Arber)

* so far > 900 Types of R.E. enzymes
have been discovered from 230 strains
of bacteria [NCERT]

R.E. enzyme

$\Rightarrow$ found only in prokaryotes (bacteria)

> These enzymes are the part of defence system of bacteria and protect the bacteria from external viral DNA.

> These enzymes never cuts the DNA of its own bacteria cz bacteria modifies the restriction site of its DNA by methylation which is done by a enzyme methylase enz

3 types of RE enzyme:

Type I, Type II and Type III

Type-I

- Not so specific
- cuts only one strand of DNA

Requirement

Mg^{+2} , ATP,

~~S-Meth~~

S-Adenyl Methionine

- Bifunctional
some enzymes have restriction and methylation activities

Type-II

v.v. specific

cuts both strands of DNA

Requirement

only Mg^{+2}

separate form for restriction and methylation

$\therefore$ commonly used.

Type-III

Type II

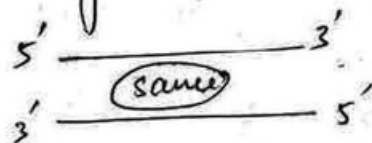
Type II

type I

type I

Restriction sites:

4/8/6 bp long special palindromic sequence

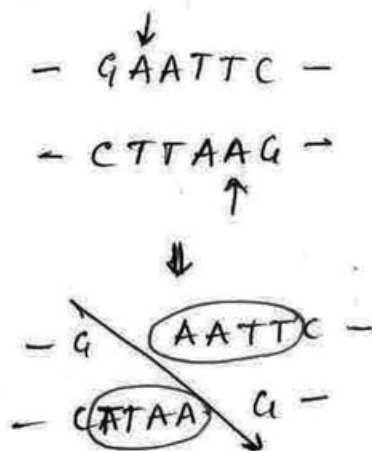


Mode of cutting

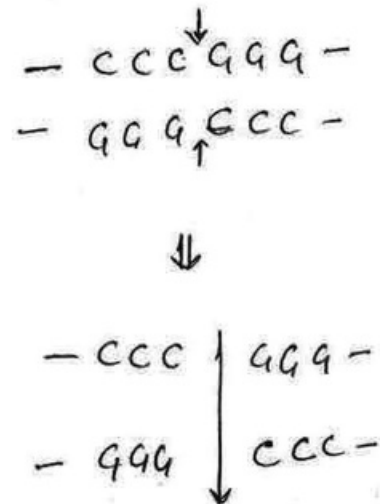
Oblique (NCERT)
cuts DNA near to the
centre of palindrone.
Produces free / sticky ends

straight
cuts DNA at the
centre of Palindrone
Produces Blunt ends.

E. coli



Sma-I



R.E enzyme

nomenclature

names of RE enzymes consist of 3/4 letters +

Roman no.

1st letter

1st letter
of bacterial
genus

2nd letter
and 3rd
letter

1st and
2nd letter
of bacterial
species

4th letter
(not essential)

↓
from
bacterial
strain

Indicates the
sequence of
Isolation
of enzyme
from bacteria.
≠ discovery

ECOR I enzyme from E. coli RI 3 Bacteria

iii) Synthesizing enzyme

enzyme used for synthesis of DNA

eg:) RNA polymerase (DNA $\rightarrow$ RNA).

Reverse transcriptase (RNA $\rightarrow$ DNA)

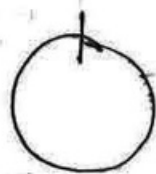
iv) Joining enzyme (molecular glue/gum)

enzyme used for joining of DNA sequence

eg; T₄ DNA ligase

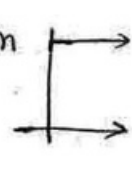
v) Other enzymes:

- > Alkaline phosphate
- > Removes PO₄ group from 5' end of DNA
- > used to prevent recircularization of any circular DNA



II) Vector DNA

- > special DNA which can carry / transfer ~~a~~ our desired gene in Host cell.
- > have self replication capacity
- * Cloning vector

Selection of vector depends on  size of desired gene
Type of host cell.

A vector must have 3 imp. properties

- i) Presence of ori - a vector DNA must have atleast one ori so that it can replicate

↓
high copy no type ($\because$ it determines the rate of replication)

- ii) Presence of selectable ~~marker~~ ^{Marker} gene -

These are the special ^{gene} which helps in identification and isolation of vector DNA/OR R DNA

eg: Antibiotic resistant gene ~~tet~~
→ tet R gene $\Rightarrow$ tetracycline R gene
→ Amp R gene $\Rightarrow$ Amphotericin R gene
enz gene $\Rightarrow$ 'gal' gene.

- iii) Presence of cloning / Restriction site?

A vector should have restriction site of several enzyme but only one site of each enzyme

Polylinker (MCS - multiple cloning sites)
of a vector have restriction site of
many enzymes

eg of vector DNA:

eg Plasmid (most common)

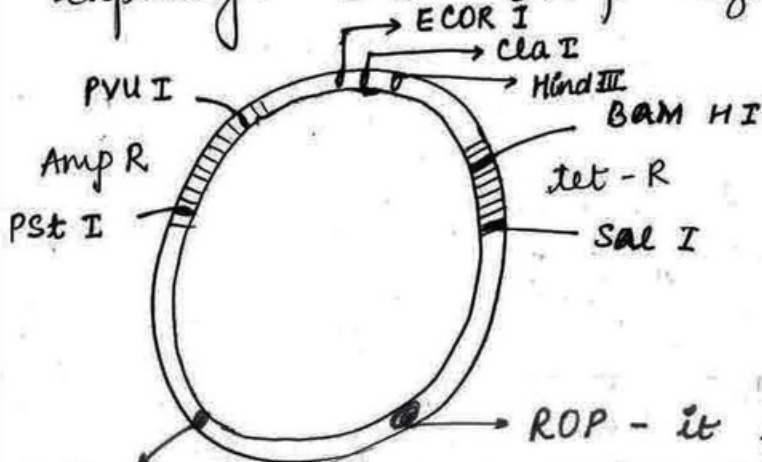
an extranuclear ds, circular, DNA present
in cytoplasm of Bacteria.

eg: $\left[\begin{array}{l} \text{pBR} 322 \\ \text{pBR} 327 \end{array} \right] \rightarrow \text{Boliver and Rodriguez}$

from E. Coli $\left[\begin{array}{l} \text{pUC} 18 \\ \text{pUC} 19 \end{array} \right] \rightarrow \text{university of California}$

$\left[\begin{array}{l} \text{Ti-plasmid} \\ \text{Ri-plasmid} \end{array} \right] \rightarrow \text{from Agrobacterium}$

capacity: 0.5 - 8 Kbp sized gene



ROP - it forms a protein
which helps in replication
of plasmid

ii) Bacteriophage DNA

eg, λ - phage.

M-13 phage.

capacity: 9-23 kbp

iii) Plant and animal viruses.

SV-40 (for mouse cell)

CMV (for plant cell)

Adenovirus

Herpes virus

Retrovirus

for animal cell

iv) Cosmid

(Artificial ~~DNA~~ vector)

['cos site' of Bacteriophage DNA + Plasmid]

12 bp long seq which helps in replication and packaging of DNA

capacity: 30-45 kbp

v) Phagemid

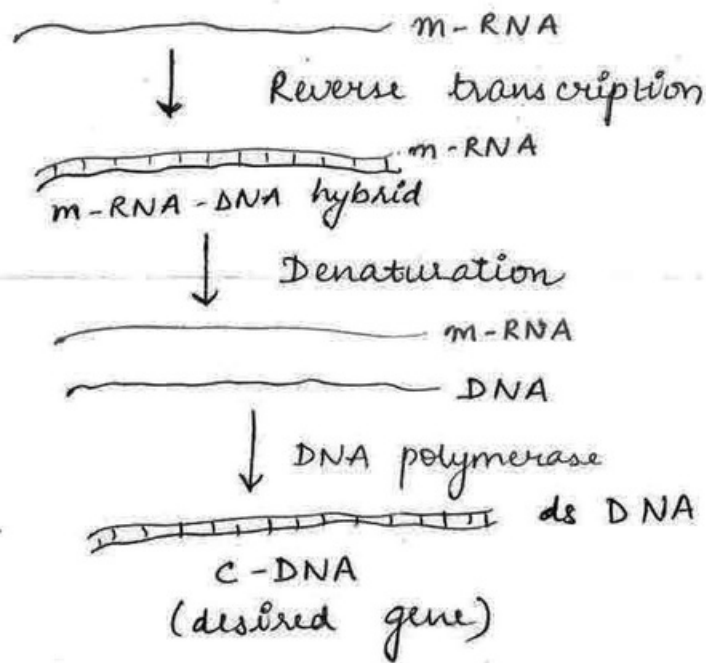
(Artificial vector)

[DNA of filamentous Bacteriophage + Plasmid]

capacity $\rightarrow$ upto 10 kbp

pUC 118

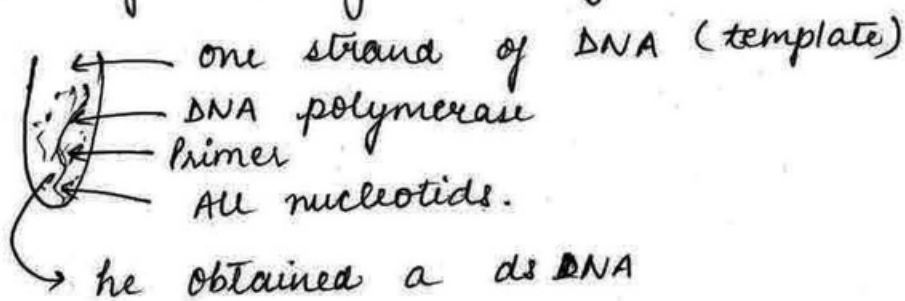
pUC 119



iii) Synthetic DNA formation

- > complete artificial synthesis of gene.
- > In this, the entire base sequence of gene must be known.

Set by ~~Kornberg~~ Kornberg (1961) [In vitro DNA replication]



Later on, H.G. Khorana → 1st complete artificial synthesis of gene (without template)

* In 1968 → he synthesised Alanine + RNA gene of yeast
77 bp long
non-functional in cell

In 1979 $\rightarrow$ he synthesised Tyrosine + RNA
gene of E. coli
207 bp long
functional in cell.

IV) Host cell: vector

- > The cell in which desired gene is to be transferred
- > several types (depends on req. of experiment)
 - Bacteria cell
 - Yeast "
 - Plant "
 - animal "
- > They are isolated from organism and their pure culture is developed.

II Procedure of r-DNA formation and gene transfer technique :

i) Isolation of DNA

cells having desired gene are taken
↓
lysis of cells by lysing enzyme
↓
Protoplasm obtained
↓
From this, some large macromolecules are removed by the use of enzyme like proteins by Protease RNA by Ribonuclease etc.
↓
Finally purified DNA can be precipitated out by chilled ethanol

ii) Cutting of DNA at specific sites to obtain desired gene / Passenger DNA

- > This desired gene / passenger DNA can be obtained by any one of the (3) methods random DNA
c-DNA
synthetic DNA
- > At the same time, a suitable vector DNA is also selected and isolated.

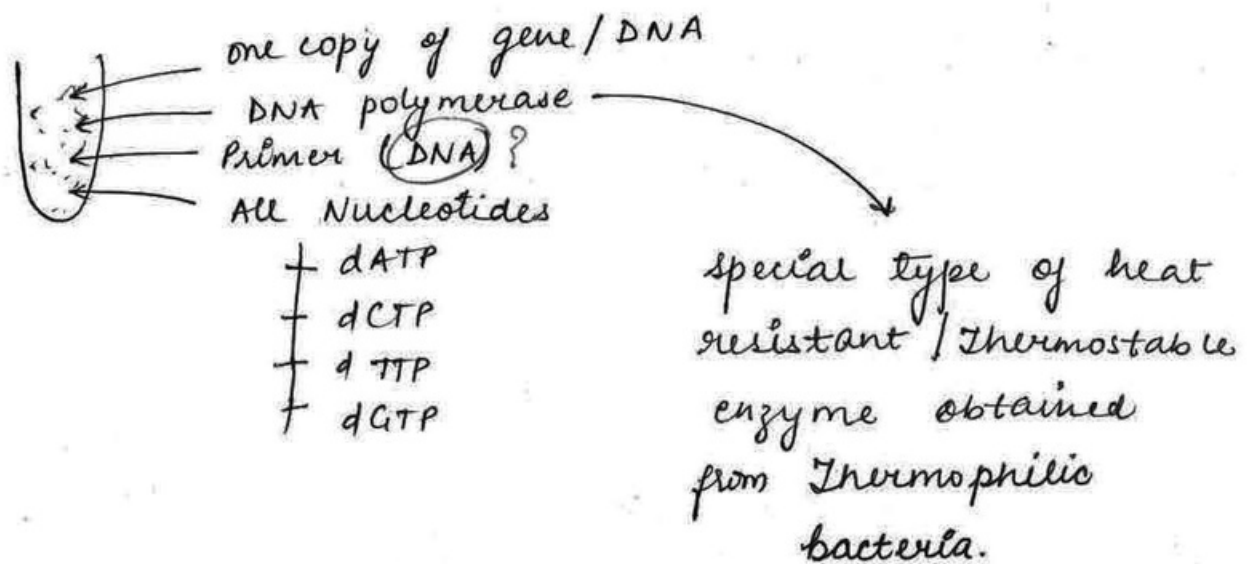
*** iii) Amplification of desired gene by PCR

The obtained desired gene is very less in amount, so amount is increased by PCR
POLYMERASE CHAIN REACTION (People's choice reaction)

PCR : It is an automated rapid technique of *in vitro* DNA Amplification.

(Reaction of 1983)

- It is based on *in vitro* DNA replication and denaturation - renaturation property of DNA



eg: Taq polymerase
from *thermus aquaticus*
(best efficiency at 72°C)

Pfu Polymerase
from *Pyrococcus furiosus*

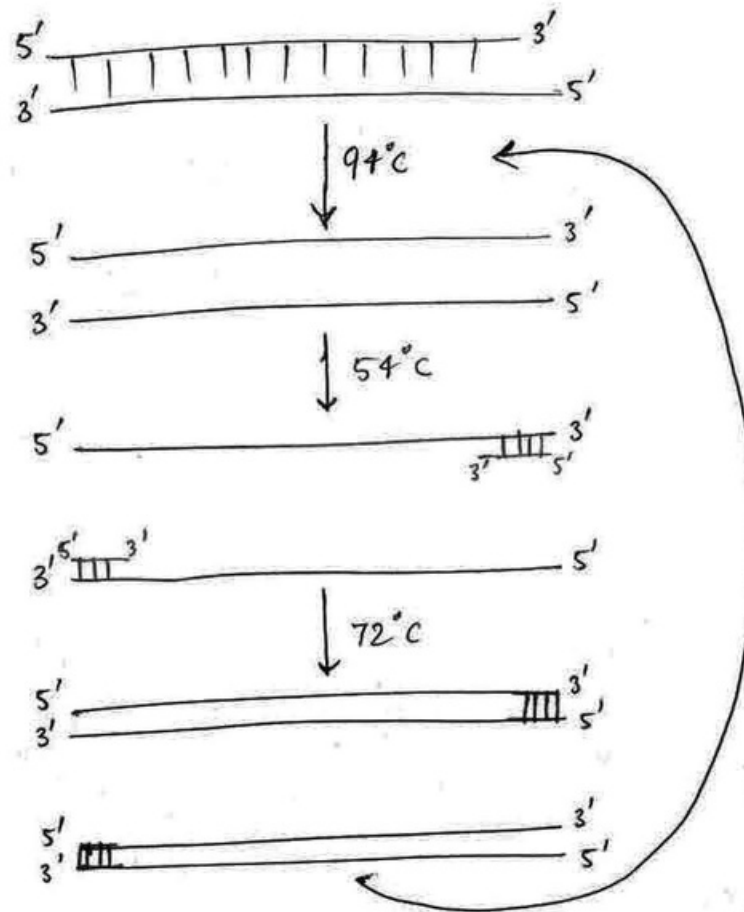
Vent Polymerase
from *Thermococcus littoralis*

PCR is a ③ step thermal cycle.

→ Denaturation (94°C)

→ Annealing (54°C)

→ Extension (72°C)



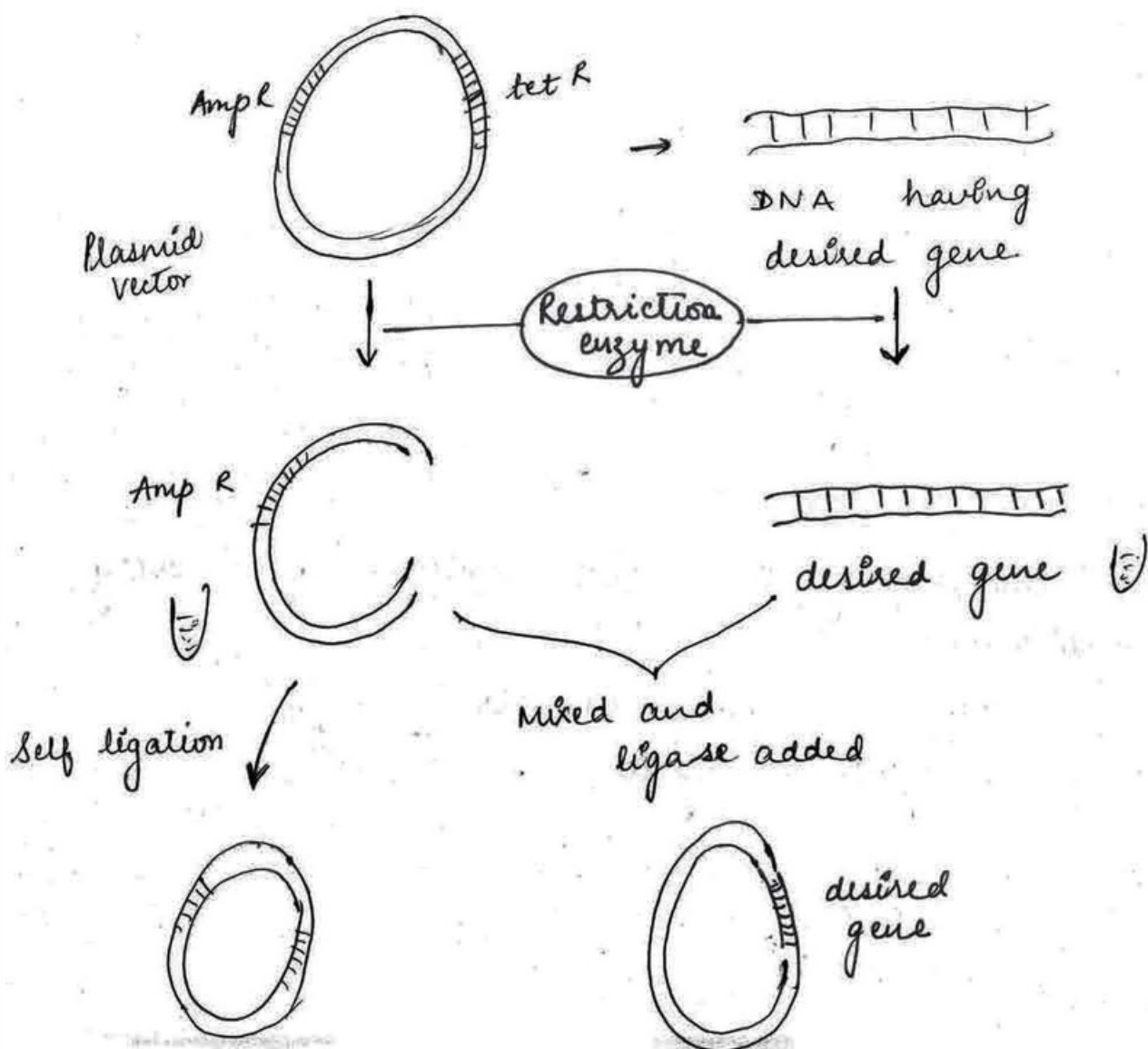
The amount of DNA gets double after every replication

No. of DNA molecule after n cycle $\Rightarrow \boxed{2^n}$

(iv) Formation of r-DNA.

> ? Both (desired gene/ passenger) DNA, ^{and vector DNA} are cut by same restriction endonuclease enzyme so that both will have complementary sticky ends. _{cut acc. to r-DNA}

> Both are mixed and ^{Phospho diester} ligase enzyme added
> ligase ~~add~~ will join passenger DNA with vector DNA and forms r-DNA



Insertional inactivation:

It is the inactivation of any marker gene of vector due to the insertion of desired gene

v) Transfer of λ -DNA in suitable host cell:

v. complicated

Several method

> depends on ~~transfer~~ ^{type} of host cells.

They are 1st made competent to take λ -DNA.

* Transfer of gene/ λ -DNA in Bacteria cell.

Bacterial cell wall are treated with lysozyme and divalent ions $[Ca^{+2}]$ calcium chloride

→ This creates pores in cell wall of bacteria and thus the efficiency of cell to take λ -DNA increases.

↓
 λ -DNA is forcefully inserted in bacterial cell by temperature/heat shock

↙ Bacterial cell and λ -DNA are placed in ice

→ suddenly shifted ~~to~~ at high temp. ($42^{\circ}C$)

→ immediately put back in ice

* Gene/κ-DNA Transfer in Plant cells.

commonly done by Ti-plasmid / Ri-plasmid vector

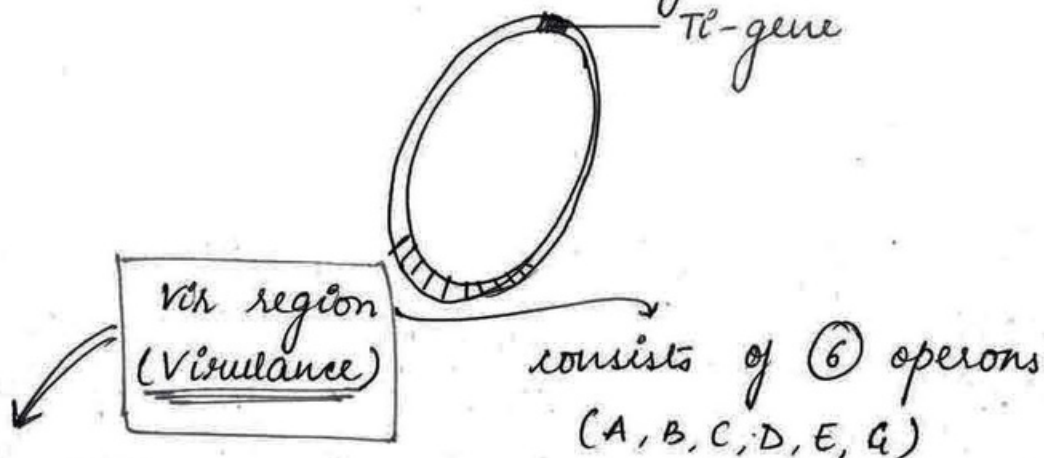
Natural genetic engineer $\Rightarrow$ form *Agrobacterium tumefaciens*

$\downarrow$
causes crown gall disease
in dicot plant
 $\neq$ monocot

$\downarrow$
form
agrobacterium
rhizogenes

$\downarrow$
causes hairy root
disease in plant.

Ti-plasmid $\Rightarrow$ Tumour inducing plasmid.



when it gets
activated, it helps
in transfer of T-DNA
(transferred DNA)
into plant genome
 $\downarrow$

Ti-gene in plant
induces tumour
formation

activated by inducer
known as Acetosyringone
(found in cells of dicot plant)
not in monocot.

Ti-gene : It is replaced by
desired gene
 $\downarrow$

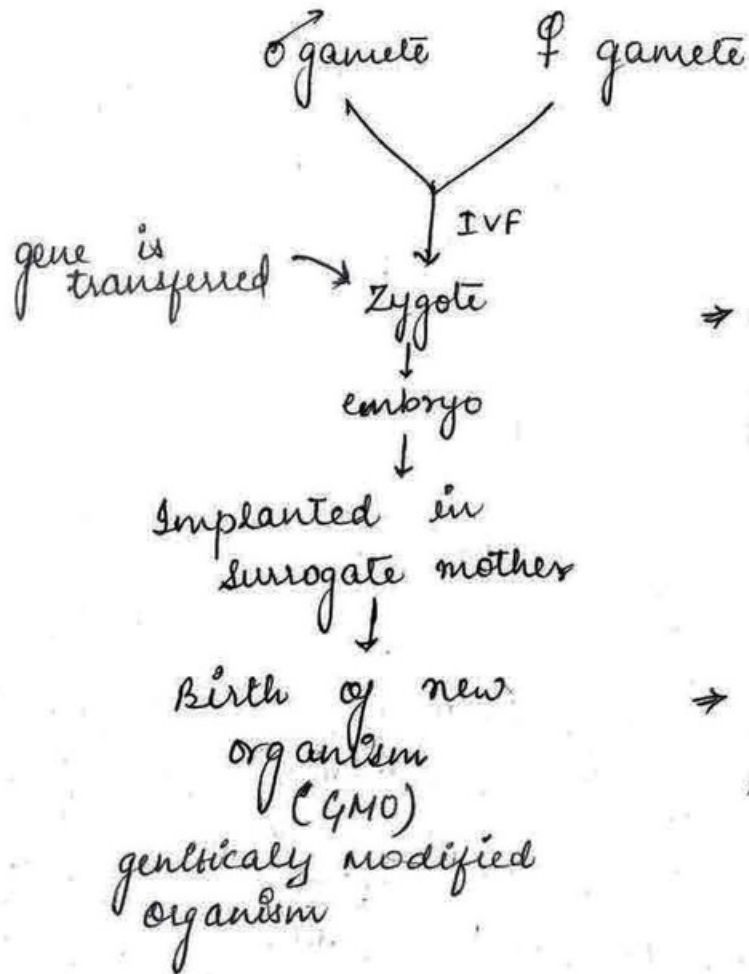
Such plasmid is known
as Disarmed plasmid

* Gene / λ -DNA Transfer in Animal.

v. v. complicated

commonly done in zygote / embryonic cell.

commonly done by some vector eg: Retrovirus
or by Vectorless gene transfer.



1st transgenic organism (animal)

⇒ A mouse in which growth hormone was inserted.

∴ size ↑↑ Supermouse

⇒ A gene of jelly fish was inserted in

Rhesus monkey

By. Gerald Schatten

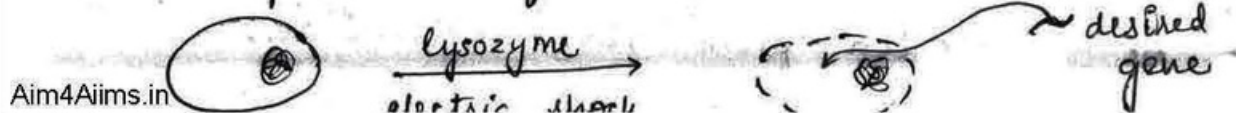
(ANDI)

[Vectorless gene transfer]:

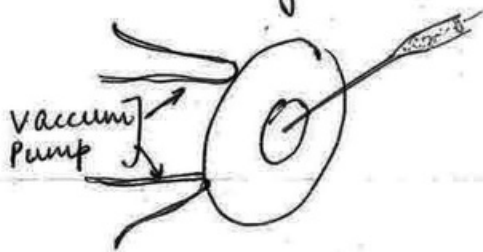
direct gene transfer

less common and have low success rate

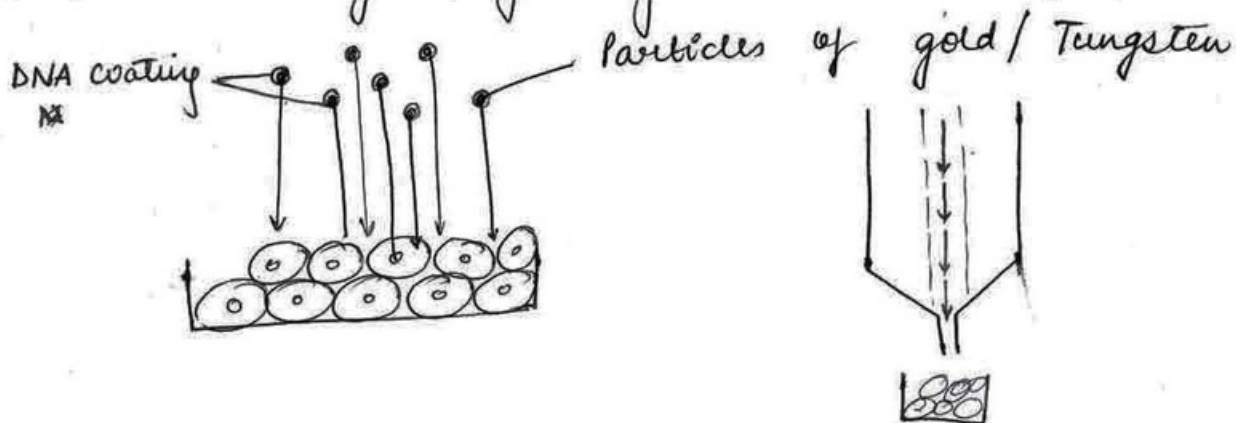
① Electroporation (for bacterial cell)



ii) Microinjection (for animal cell)

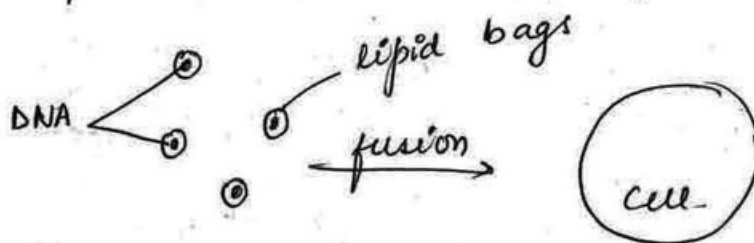


iii) Particle gun / gene gun (Biolistic) (for plants)



iv) Chemically mediated Transfer
PEG (Polyethylene glycol)

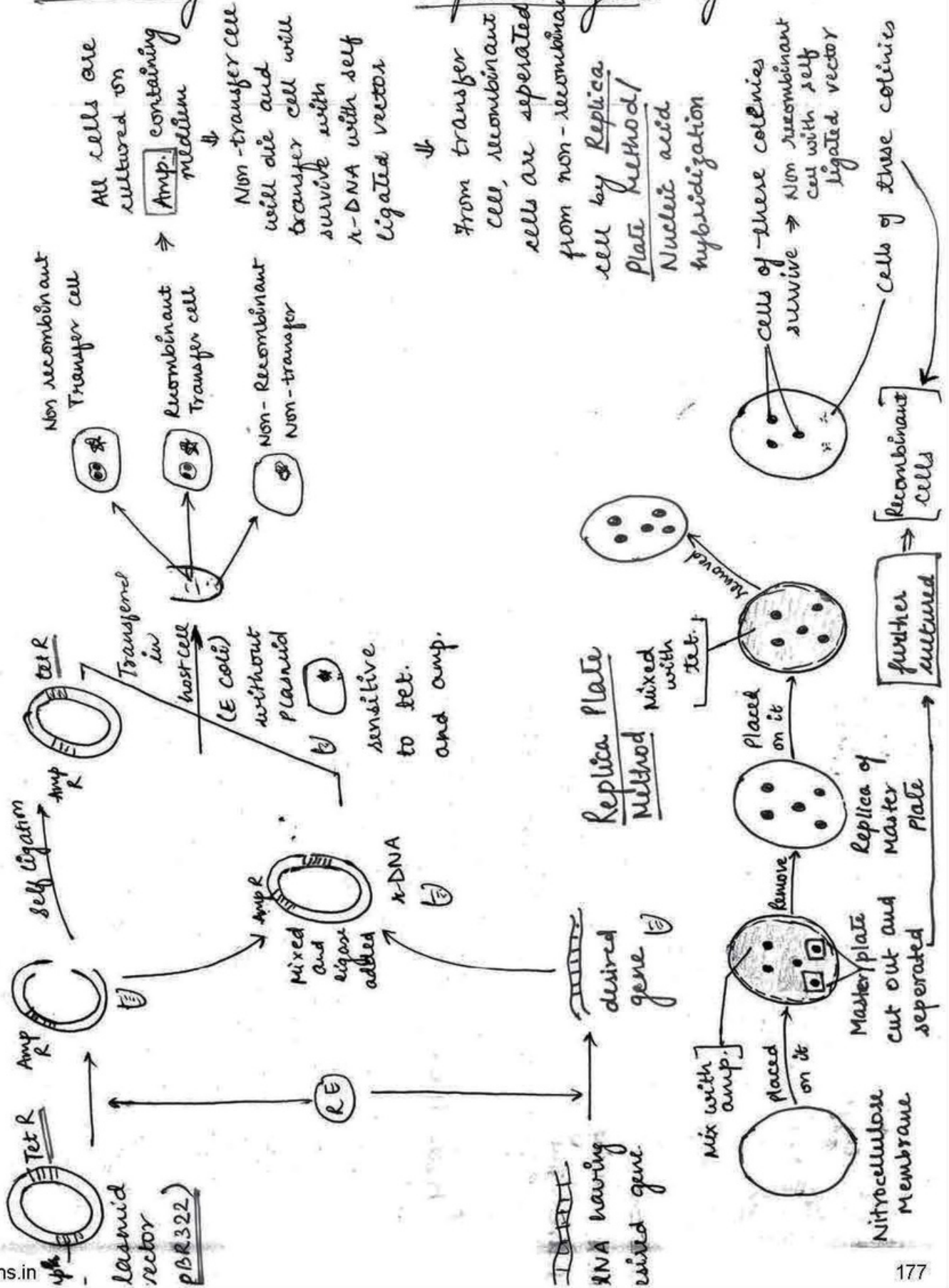
v) Liposome mediated Transfer



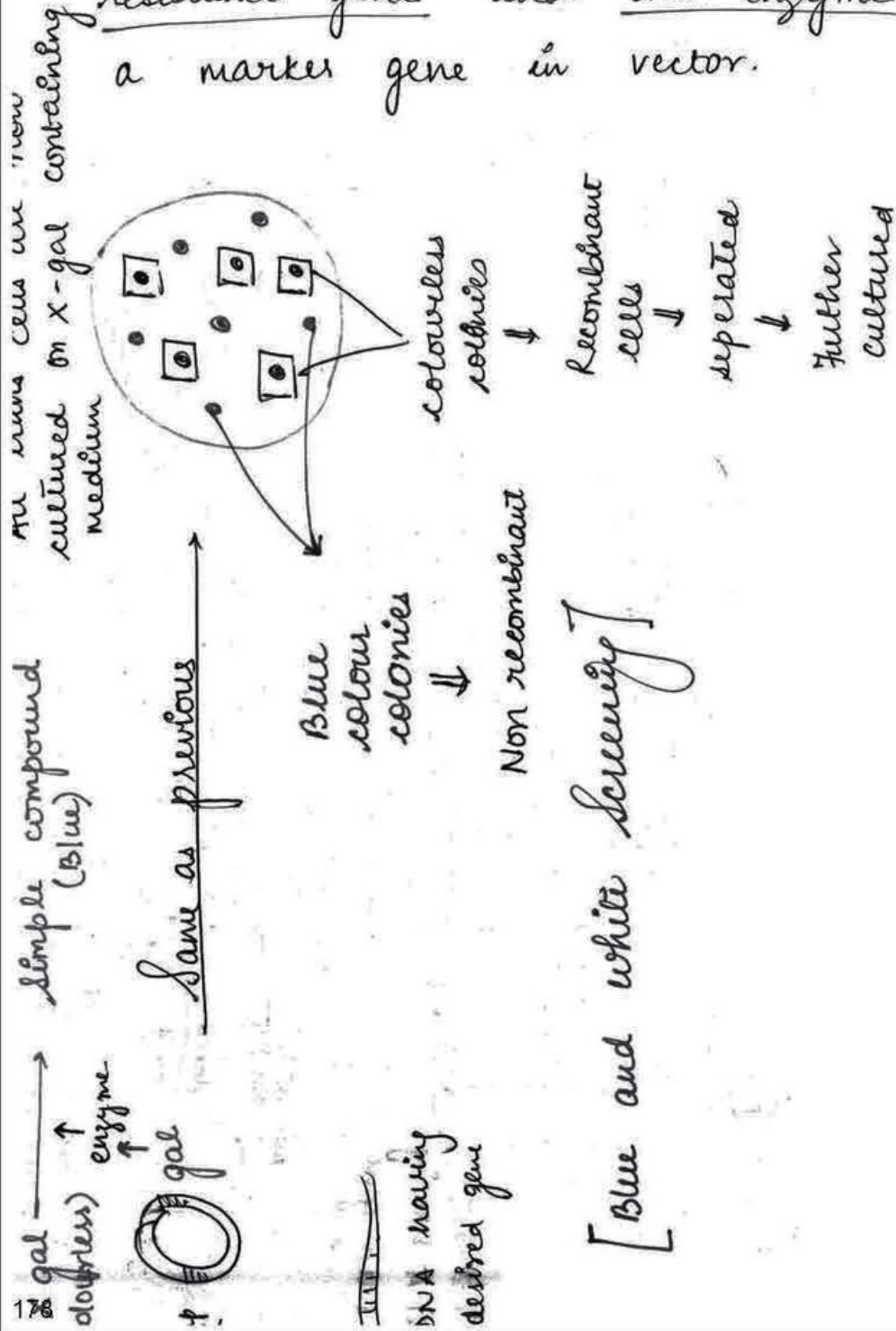
vi) Selection and screening of Recombinant host cell:

→ Recombinant DNA will not transfer in all the host cell so recombinant host cell must be separated from non-recombinant

host cell by the help of selectable
marker gene or reporter gene of vector



» When both the marker genes are antibiotic resistant gene in vector, then it is difficult to separate recombinant cell from non-recombinant having self-ligated vector. So, it is better to take one antibiotic resistance gene and one enzyme gene as a marker gene in vector.



[Blue and white screening]

vii) Obtaining desired gene product at large scale.
> The desired gene will express itself in recombinant host cell (heterologous host) and produces recombinant protein.

> The obtained recombinant host cells are very less in amount so they are now cultured at large scale so that desired product can be obtained in sufficient quantity.

Production techniques develops at ③ levels:

- Lab level [UPSTREAM PROCESS]
- Pilot plant level
- Industrial level [DOWNSTREAM PROCESS]

i) lab level: It is the 1st level of technological development, here, a fundamental technique is developed i.e. gene is transferred into the host cell. In this, cells are cultured in small amount of medium taken in glass wears. At this level all the essential growth condition necessary for culture of cells are standardise like, type of medium, pH, Temp, O_2 availability etc.

ii) Pilot plant level :

Intermediate level

At this level, cells are cultured in large amount of medium and glass wears are replaced by large metallic containers known as bioreactor

At this level, the cost of the product is also calculated.

Bioreactor : 100 - 1000 l

- > provides all essential growth condition necessary for culture of cells.
- > have two types of growth system

- | <u>support</u> | <u>suspended</u> |
|---|---|
| > solid medium
(cell cultured) | liquid medium |
| > A continuous / agitation
of medium for aeration
is not required | A continuous / agitation
of medium for
aeration is required |
| > difficult to transport | easier to transfer. |

have two types of culture system

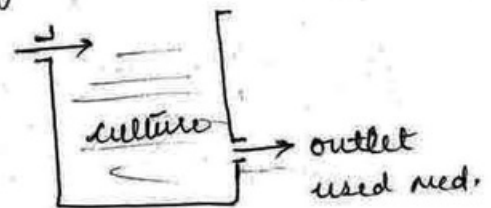
have two types of culture system.

Batch

- > cells are cultured in batches of medium
- > here, both types of medium can be used

continuous

- > In this a continuous system is developed where fresh medium is added from one side and used medium is taken out by other sides and in culture, cells are maintained as physiologically most active log or exponential phase of growth.



commonly stirring types of Bioreactor are used

simple stirred tank type

sparged stirred tank type

iii) Industrial level: At this level, cells are cultured in very large amount of medium taken in large sized bioreactor and a permanent manufacturing unit is started.

Downstream processing:

starts after the completion of bio-synthetic phase

» The product obtained in bioreactor undergoes several processing and finally it is converted into usable form, all these processings are collectively known as downstream processing.

→ Separation and purification of products

→ Addition of preservatives etc.

→ Quality control testing

→ Clinical trials (if medicine)

* ~~As~~ Downstream processings are different for different products.

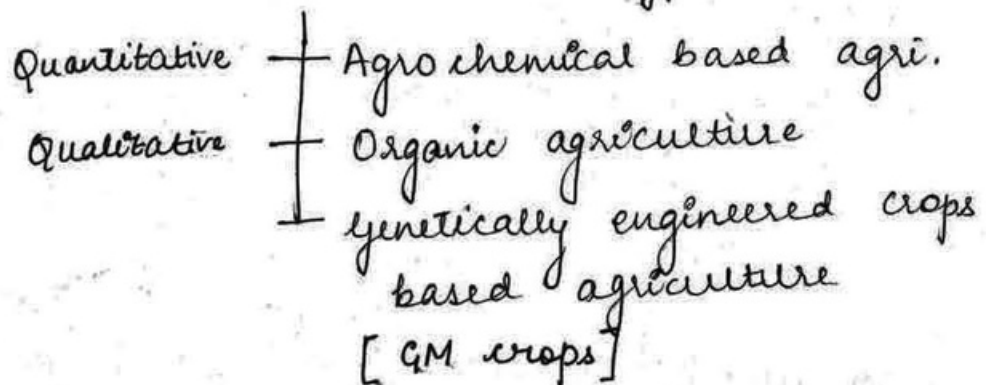
APPLICATION OF BIOTECHNOLOGY

(λ -DNA Tech. & gene transfer tech)

In several fields : agricultural / food science
medical science / Biopharmaceutical
environment control /
other fields

- green biotech (agricultural)
- Red biotech (Medical science)
- white / grey biotech (Industrial)
- Blue biotech (marine science)

Application of biotech in agriculture :
↓
③ types



GM crops are produced:

i) to enhance nutritional quality of food

eg: golden rice

Vit A rich (carotene gene was transferred)

Protein rich potato

ii) to develop tolerance against abiotic stresses of flood, Drought, frost, salinity etc.

eg: Salt Tolerant transgenic Tomato.

In this, a gene was transferred which highly express $\text{Na}^+ \text{P}^+$ antiport pump in its cell membrane by which it accumulates all additional ~~ion~~ ions in its vacuoles and later on release them out.

iii) To reduce post harvest losses

eg: Flavr Savr Tomato *

have long shelf life due to delayed ripening.

In, this, gene of ripening enzyme, Polygalacturonase was inhibited by the use of Antisense Technology (gene of coding strand - dsRNA)

iv) to develop herbicide resistant plants

eg: Herbicide resistant Transgenic Tobacco
(1st tobacco plant)

resistant to sulphonyl urea and glyphosphate

In this tobacco ALS gene of Arabidopsis was transferred
(Acetolactate synthetase)

v) For production of Biopharmaceutical compound.

eg: A gene of Hirudin is transferred in
Brassica napus (Rapeseed oil)

⇒ It produces hirudin and collects in its seeds
It is crushed → oil obtained → Hirudin separated
(Blood Anticoagulant)

* Edible vaccine

Transgenic banana } for hepatitis and cholera
" Tomato }

" Sugarbeet → for foot and mouth disease

Banana } for diarrhoea
Potato }

★★ vi) For development of Pest Resistant Plants

a) ⇒ Insect resistant Plant

For this purpose (Bt) plants/crops have been
developed
↓
Bacillus thuringiensis

> In this plant Bt gene/cry gene of Bacillus
thuringiensis Bacteria was inserted.

> Bt gene/cry gene → It forms an insecticidal/toxic
protein which kills many insects

↓
Lepidopteran
- Tobacco budworm

↓
Coleopteron
- Beetles

↓
Dipteran
- Flies

Bacteria will not be killed by this protein because it is found in inactive insoluble crystal form $\Rightarrow$ crystal protein
(Cry protein)

* when insect feed this protein, the crystals of this protein gets dissolved and becomes activated in the midgut of insect due to Alkaline medium

> This protein now binds with midgut epithelium, creates pores and causes cell swelling and lysis and ultimately the insect will die.

Bt gene/cry gene are of several types:
(Insect group specific)

Selection of gene depends on two things

- i) Targeted insect
- ii) host plant

eg: cry I Ac and cry II Ab gene $\Rightarrow$ for bollworm
cry I Ab gene $\Rightarrow$ for corn borer insect

* This gene is isolated from bacteria and transferred in many plant with the help of Ti plasmid.

eg: Bt cotton (commonly used)

→ Resistant to Bollworm

Bt ~~sugar~~ soyabean

Bt Tobacco

Bt corn

Bt rice

Bt Brinjal

* Bt genes are avoided, because they are toxic in food plant

b) → Nematode resistant plants.

Some causes disease in plants and animals

Meloidogyne Incognita

causes root knot disease plant (Tobacco)

→ developed by a unique technique known as RNAi (RNA interference) technique.

↓
Technique of gene silencing

detailed study by → Fire and mello
(2006 - Nobel prize)

In this the m-RNA of any targeted gene is destroyed by complementary double stranded

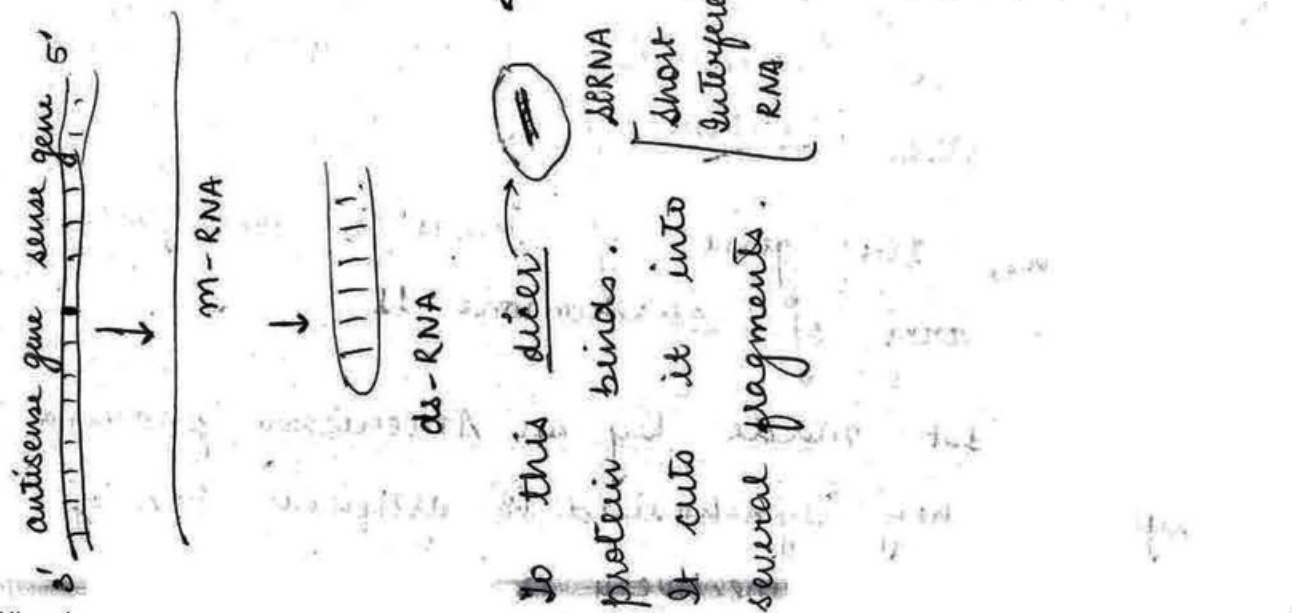
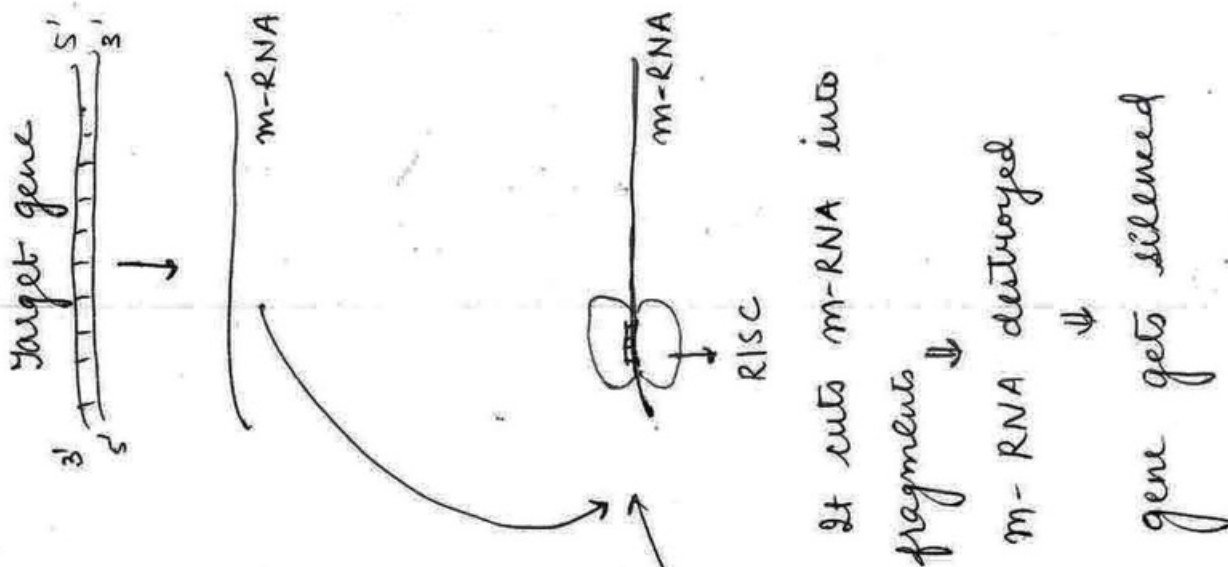
RNA

source of double stranded RNA

It can be incorporated in cells with the help of vectors like transposone and retro virus

OR A sense and anti-sense gene of targeted gene is transferred in the cells which produces a double stranded RNA inside the cell and this dsRNA will initiate RNA interference and results in gene silencing.

* For nematode resistant, a sense and antisense gene of nematode specific gene are introduced in plant cells, which forms a dsRNA in plant cell, when nematode infection takes place, this dsRNA of plant cell destroy the m-RNA of ~~the~~ nematode due to which nematode is unable ~~to~~ to form a specific protein and ultimately it dies and plant becomes nematode resistance.



Application of Biotech in Medical science formations of medicines → known as recombinant Therapeutics

(Pure, safe, more effective,
have less side effect)

so far about (30) recombinant medicines are
used all over world

In India, only (12) are used

Diagnosis of disease

Treatment of disease

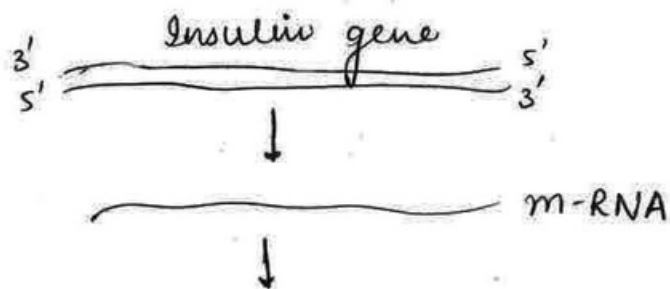
* a) Genetically engineered Insulin
a proteinous hormone
regulates blood-glucose level

If it is less in amount → diabetes occurs

In such conditions, it is given from outside
Initially it was obtained from cattles and
pigs (Bovine insuline)

In humans, the gene of insulin is present
on short arm of chromosome 11

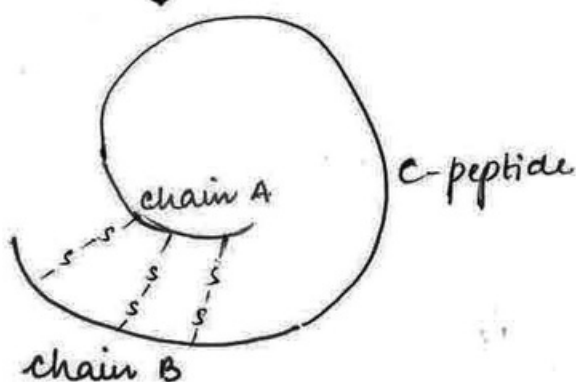
* It was 1st made by an American pharma
company, They synthesised 2 artificial DNA of
chain A and B respectively and transferred in



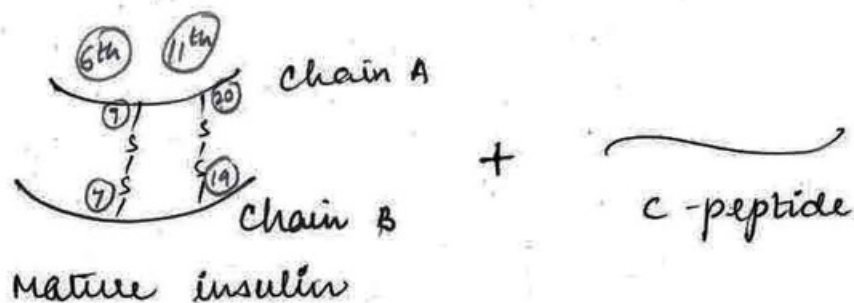
chain A (21 AA) C-peptide (33 AA) chain B (30 AA)

Proinsulin

↓ formation of disulphide bonds



↓ Removal of C-peptide (by β -cell peptides)



Artificial DNA
of chain A

↓ by plasmid vector
in E. coli

→ chain A obtained

→ chain B obtained
in E. coli

↑ by plasmid vector

Artificial DNA
of chain B

They are joined by
disulphide bonds.

↓
Insulin forms

↓
'Humulin'
(5th July 1983)

Humalog was more purified by → Eli Lilly

↳ Ramboxy
(Indian
Pharma
Company)

b) Gene Therapy:

Technique for treatment of genetical disorder

> ~~tot~~ It is the collection of all those methods which are involved in ~~collection~~ correction of a gene defect that has been identified in an embryo or a child.

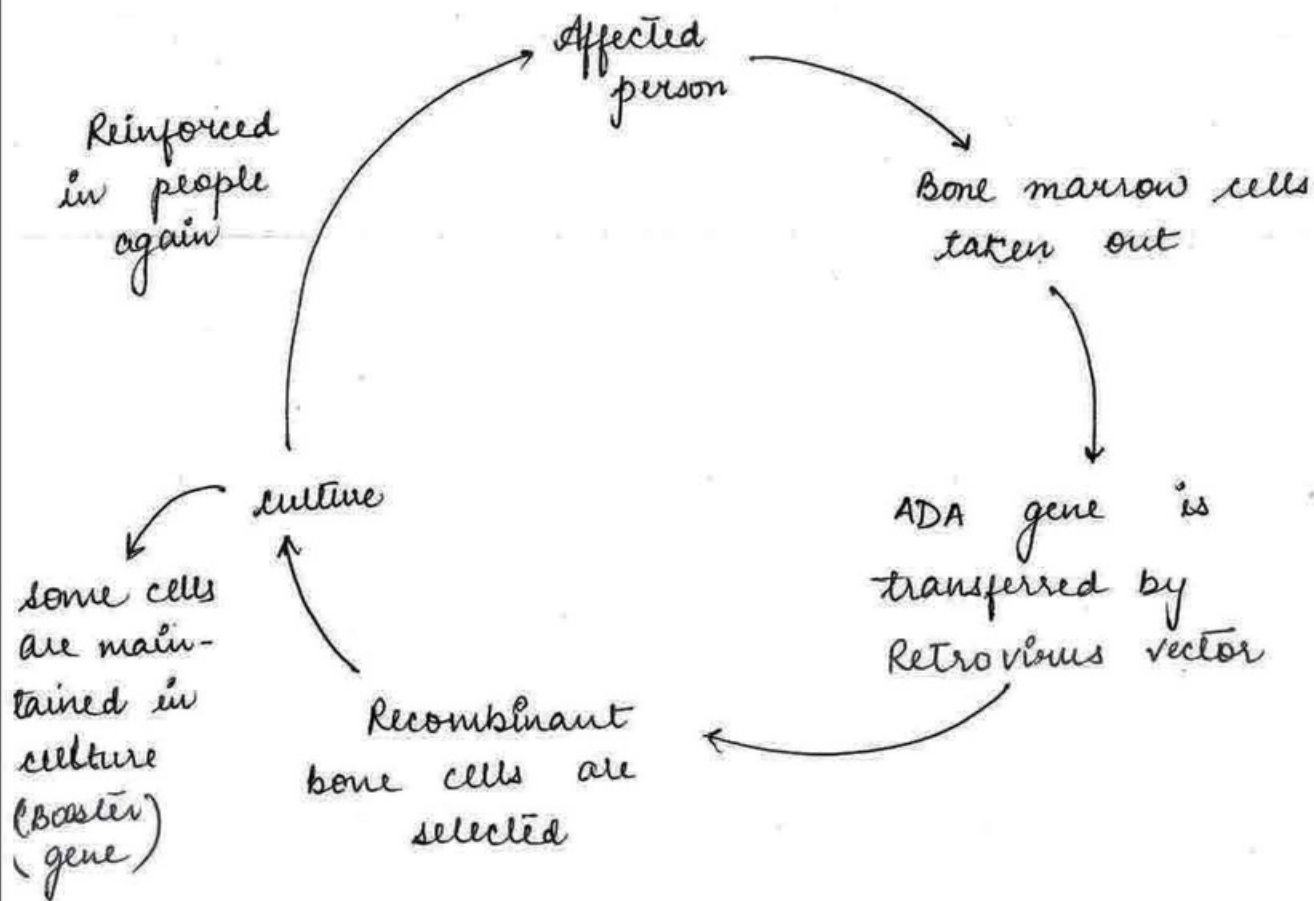
* 1st successful gene therapy was done in 1990 in a 4 year girl, suffering from ADA^{enzyme} deficiency

↓
It results in SCID
(severe combined
immuno deficiency)

Adenosine deaminase
↓
it is required for
the formation of
functional T lymphocyte
from bone marrow cell

Treatment:

- i) enzyme replacement therapy
- ii) Bone marrow transplantation
- iii) gene therapy



» For permanent cure, then gene transfer should be ~~ago~~ done at embryonic stage

c) Molecular Diagnosis:

The techniques of biotech or genetic engineering are used to find out any infection/pathogen very accurately ~~at~~ and very early stage

It may involve:

PCR

ELISA

Transgenic animals:

- > Several animals have been produced
- > animals ~~have been~~ in which genes have been transferred

eg: Mouse $\Rightarrow$ 95%

sheep

goat

cow

pigs etc

uses and applications of transgenic animals:

- > For production of biologically / industrially important industrial products.

<u>Organs</u>	<u>gene transferred</u>	<u>uses</u>
i) Transgenic sheep	Human blood clotting factor VII and IX gene	for the treatment of Haemophilia.
ii) Transgenic sheep	Human α antitrypsin gene	for the treatment of Emphysema
iii) Transgenic cow	Human Lactoferrin gene	for the treatment of Cystic fibrosis

- | | | |
|--|--|---|
| iv) Transgenic cow
ROSLIE | Human lactalbumin gene | produces nutritious milk used for feeding infants
(2.4 g/lit lactalbumin) |
| v) Transgenic goat | TPA gene/
streptokinase
(Tissue plasminogen activator) | used for dissolution of blood clots in in heart and blood vessels. |
| vi) Transgenic goat | spider gene | (forms thread) used for formation of cloths and other products. |
| vii) Transgenic pigs | Human antigen gene | Its organs are used for transplantation. |
| viii) Transgenic pig / sheep / goat / cow / fish etc | hGH | (size ↑↑es)
can be used as food (GMF)
genetically modified food |

» For the study of normal physiology and development

(study of gene expression and regulation)

Transgenic mice

» For the study of disease

eg: Oncomouse (for cancer study)

» For vaccine safety testing

Transgenic mice are used for safety testing of Polio-vaccine

» chemical safety / toxicity testing

2. For this purpose, sensitive transgenic organisms are developed which are used for safety testing.

d) Transgenic microbes

max. Organism

eg: Transgenic E. coli

Chorionic gonado-
trophin

(for treatment of
sterility / infertility)

having human insulin gene
(for diabetes treatment);

hGH gene (dwarfism treat.)

calcitonin gene (rickets tre.)

Interferon gene
(for viral resistance)

Antigen gene

Pseudomonas putida (superbug)

have ability to digest hydrocarbons (petroleum)

use: to remove oil spills from oceans

In this 4 types of genes/plasmid was transferred
OCT, XYL, CAM, NAH